



کلمہ الہی
ذکر و تالیف

LECTURE - (3) TRANSFER FUNCTION

TRANSFER FUNCTION Outline

Outline

- **Linear Ordinary Differential Equations.**
- **Solution of Linear ODEs and Laplace Transform**
- **Laplace Transform**
- **Transfer function of linear ODES**
- **Some common transfer functions**
- **Poles and Zeros of Transfer Function**

- In general, the differential equation of an nth-order system is written as:

$$\frac{dy^{(n)}(t)}{dt^n} + a_{n-1} \frac{dy^{(n-1)}(t)}{dt^{n-1}} + \dots + a_1 \dot{y}(t) + a_0 y(t) = b_{n-1} \frac{du^{(n-1)}(t)}{dt} + b_{n-2} \frac{du^{(n-2)}(t)}{dt} + \dots + b_1 \dot{u}(t) + b_0 u(t)$$

- which is also known as a **linear ordinary differential equation** if the coefficients $a_0, a_1, \dots, a_{n-1}, a_n$ and $b_0, b_1, \dots, b_{n-1}$ are real constants and are not functions of $y(t)$.
- In control systems, the terms $u(t)$ and $y(t)$ are known as **the input and the output** of the system, respectively, for $t \geq t_0$.
- Prototype second-order system:**

$$\ddot{y}(t) + 2\zeta\omega_n \dot{y}(t) + \omega_n^2 y(t) = \omega_n^2 r(t)$$

- where ω and ζ are the natural frequency and the damping ratio of the system, respectively. We define $y(t)$ as **the output** and $r(t)$ as **the input** of the system.



- Once we obtain the equations of the system, we need to develop a set of analytical and numerical tools that can assist us with a clear understanding of the performance of the system.
- This is an important step in advance of extending the design of a control system to a prototype or the actual system.
- Two most common tools for studying the behavior of (i.e., the solution of) a control system are the transfer function.
- Transfer functions are based on the Laplace transform technique and are valid only for linear time-invariant systems.
- Laplace transform technique is one of the mathematical tools used to solve linear ordinary differential equations.
- This approach is very popular in the study of control systems because of the following two features:
 - The homogeneous and the particular components in the solution of the differential equation are obtained in one operation.
 - The Laplace transform transforms the differential equation into a simple to manipulate algebraic form—in what is known as the s-domain



- Laplace Transform**

- Consider a function $f(t)$, $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(t) = 0$ for all $t < 0$.



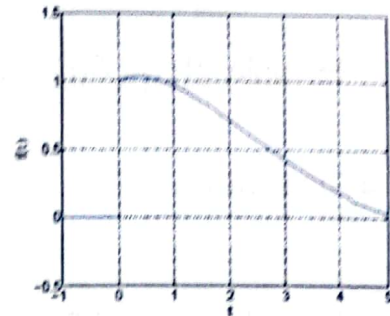
Pierre-Simon Laplace
(1749-1827)

Definition

The **Laplace transform** $\mathcal{L}[f]$ of f is the function $F: \mathbb{C} \rightarrow \mathbb{C}$ of complex variable $s \in \mathbb{C}$ defined by

$$F(s) = \int_0^{+\infty} e^{-st} f(t) dt$$

for all $s \in \mathbb{C}$ for which the integral exists



- The variable s is referred to as the Laplace operator, which is a complex variable; that is, $s = \sigma + j\omega$, where σ is the real component, $j = \sqrt{-1}$, ω the imaginary component,
- Laplace transforms convert integral and differential equations into algebraic equations. We'll see how ...



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- Examples of Laplace transforms:**

- Unit step**

$$f(t) = \mathbb{I}(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases} \Rightarrow F(s) = \int_0^{+\infty} e^{-st} dt = -\frac{1}{s} \Big|_0^{+\infty} = \frac{1}{s}$$

- Dirac's delta (or impulse function¹)**

$$f(t) = \delta(t) \triangleq \begin{cases} 0 & \text{if } t \neq 0 \\ +\infty & \text{if } t = 0 \end{cases} \text{ such that } \int_{-\infty}^{+\infty} \delta(t) dt = 1$$

$$F(s) = \int_0^{+\infty} \delta(t) e^{-st} dt = e^{-s \cdot 0} = 1, \forall s \in \mathbb{C}$$

- Exponential function:** $f(t) = e^{-at}$ $t \geq 0$ where a is a real constant.

Laplace transform of $f(t)$:

$$F(s) = \int_0^{\infty} e^{-at} e^{-st} dt = \frac{e^{-(s+a)t}}{s+a} \Big|_0^{\infty} = \frac{1}{s+a}$$



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• **Properties of Laplace transforms:**

• **Linearity**

$$\mathcal{L}[k_1 f_1(t) + k_2 f_2(t)] = k_1 \mathcal{L}[f_1(t)] + k_2 \mathcal{L}[f_2(t)]$$

Example: $f(t) = \delta(t) - 2 \mathbb{I}(t) \Rightarrow \mathcal{L}[f] = 1 - \frac{2}{s}$

• **Time delay**

$$\mathcal{L}[f(t - \tau)] = e^{-s\tau} \mathcal{L}[f(t)]$$

Example: $f(t) = 3 \mathbb{I}(t - 2) \Rightarrow \mathcal{L}[f] = \frac{3e^{-2s}}{s}$

• **Exponential scaling**

$$\mathcal{L}[e^{at} f(t)] = F(s - a), \text{ where } F(s) = \mathcal{L}[f(t)]$$

Example: $f(t) = e^{at} \mathbb{I}(t) \Rightarrow \mathcal{L}[f] = \frac{1}{s - a}$

Example: $f(t) = \cos(\omega t) \mathbb{I}(t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2} \mathbb{I}(t) \Rightarrow \mathcal{L}[f] = \frac{s}{s^2 + \omega^2}$



• **Properties of Laplace transforms:**

• **Time derivative²:**

$$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = s\mathcal{L}[f(t)] - f(0^+)$$

$f(0^+) = \lim_{t \rightarrow 0^+} f(t)$. If f is continuous in 0, $f(0^+) = f(0)$

Example $\Rightarrow f(t) = \sin(\omega t) \mathbb{I}(t) \Rightarrow \mathcal{L}[f] = \frac{\omega}{s^2 + \omega^2}$

• **Multiplication by t**

$$\mathcal{L}[tf(t)] = -\frac{d}{ds} \mathcal{L}[f(t)]$$

Example $\Rightarrow f(t) = t \mathbb{I}(t) \Rightarrow \mathcal{L}[f] = \frac{1}{s^2}$



• Initial and final value theorems:

Initial value theorem

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s)$$

Example: $f(t) = \mathbb{I}(t) - t \mathbb{I}(t) \Rightarrow F(s) = \frac{1}{s} - \frac{1}{s^2}$
 $f(0^+) = 1 = \lim_{s \rightarrow \infty} sF(s)$

Final value theorem

$$\lim_{t \rightarrow +\infty} f(t) = \lim_{s \rightarrow 0} sF(s)$$

Example: $f(t) = \mathbb{I}(t) - e^{-t} \mathbb{I}(t) \Rightarrow F(s) = \frac{1}{s} - \frac{1}{s+1}$
 $f(+\infty) = 1 = \lim_{s \rightarrow 0} sF(s)$

• Common Laplace transforms:

1	$\frac{1}{s}$
δ	1
$\delta^{(k)}$	s^k
t	$\frac{1}{s^2}$
$\frac{t^k}{k!}, k \geq 0$	$\frac{1}{s^{k+1}}$
e^{at}	$\frac{1}{s-a}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2} = \frac{1/2}{s-j\omega} + \frac{1/2}{s+j\omega}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2} = \frac{1/2j}{s-j\omega} - \frac{1/2j}{s+j\omega}$
$\cos(\omega t + \phi)$	$\frac{s \cos \phi - \omega \sin \phi}{s^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$

In MATLAB use

$F = \text{LAPLACE}(f)$

MATLAB

```
» syms t
» f=exp(2*t)+t-t^2
» F=laplace(f)
```

F =

$1/(s-2) + 1/s^2 - 2/s^3$

Properties of Laplace transforms

Properties

$$f(t)$$

$$f + g$$

$$\alpha f \quad (\alpha \in \mathbb{R})$$

$$\frac{df}{dt}$$

$$\frac{d^k f}{dt^k}$$

$$g(t) = \int_0^t f(\tau) d\tau$$

$$f(\alpha t), \alpha > 0$$

$$e^{at} f(t)$$

$$t f(t)$$

$$t^k f(t)$$

$$\frac{f(t)}{t}$$

$$g(t) = \begin{cases} 0 & 0 \leq t < T \\ f(t-T) & t \geq T \end{cases}, T \geq 0 \quad G(s) = e^{-sT} F(s)$$

$$\alpha F$$

$$sF(s) - f(0)$$

$$s^k F(s) - s^{k-1} f(0) - s^{k-2} \frac{df}{dt}(0) - \dots - \frac{d^{k-1} f}{dt^{k-1}}(0)$$

$$G(s) = \frac{F(s)}{s}$$

$$\frac{1}{\alpha} F(s/\alpha)$$

$$F(s-a)$$

$$-\frac{dF}{ds}$$

$$(-1)^k \frac{d^k F(s)}{ds^k}$$

$$\int_s^\infty F(s) ds$$

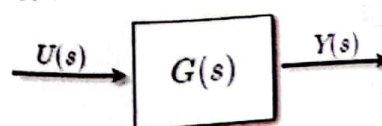


TRANSFER FUNCTION Transfer Function of linear ODES

- Transfer Function of Linear ODES
- In classical control, transfer functions are used to represent input-output relations between variables.
- Let us consider the following n th-order differential equation with constant real coefficients:

$$\frac{dy^{(n)}(t)}{dt^n} + a_{n-1} \frac{dy^{(n-1)}(t)}{dt^{n-1}} + \dots + a_1 \dot{y}(t) + a_0 y(t) = b_m \frac{du^{(m)}(t)}{dt^m} + b_{m-1} \frac{du^{(m-1)}(t)}{dt^{m-1}} + \dots + b_1 \dot{u}(t) + b_0 u(t)$$

- The transfer function between a pair of input ($u(t)$) and output ($y(t)$) variables is the ratio of the Laplace transform of the output ($Y(s)$) to the Laplace transform of the input ($U(s)$).



• Transfer Function of Linear ODES

- Taking the Laplace transform on both sides of the equation and assume zero initial conditions $y(0) = \dot{y}(0) = y^{(n-1)}(0) = 0$. The result is:

$$(s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0)Y(s) = (b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0)U(s)$$

- we obtain immediately the transfer function from u to y :

$$G(s) = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0} \xrightarrow{U(s)} \boxed{G(s)} \xrightarrow{Y(s)}$$

Example

$$\ddot{y} + 11\dot{y} + 10y = 2\dot{u} + 22u$$

$$G(s) = \frac{2s + 22}{s^2 + 11s + 10}$$

```

MATLAB
»G=tf([2 22],[1 11 10])

Transfer function:
 2 s + 22
-----
 s^2 + 11 s + 10
    
```

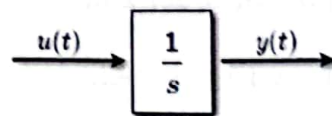
TRANSFER FUNCTION Some common transfer functions

• Some common transfer functions

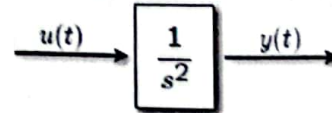
- Differential equation

$$\ddot{y}(t) + 3\dot{y}(t) + y(t) = \dot{u}(t) + u(t) \quad \bullet \text{ The transfer function is } G(s) = \frac{s+1}{s^2+3s+1}$$

$$y(t) = \int_0^t u(\tau) d\tau$$



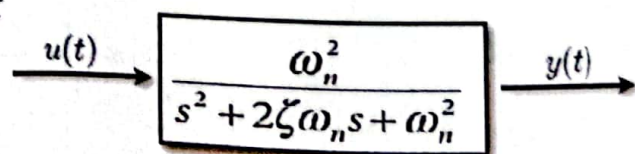
$$y(t) = \iint_0^t u(\tau) d\tau$$



- Damped oscillator with frequency ω_n rad/s and damping factor ζ

$$\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2y(t) = \omega_n^2r(t)$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_ns + \omega_n^2}$$



- **Poles and Zeros:**



- Rewrite the transfer function as the ratio of polynomials ($m < n$):

$$G(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0} = \frac{N(s)}{D(s)}$$

- The roots p_i of $D(s)$ are called the poles of the linear system $G(s)$
- The roots z_i of $N(s)$ are called the zeros of $G(s)$
- $G(s)$ is often written in zero/pole/gain form:

$$G(s) = K \frac{(s - z_1) \dots (s - z_m)}{(s - p_1) \dots (s - p_n)}$$

(In MATLAB use **ZPK** to transform to zero/pole/gain form)



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TRANSFER FUNCTION

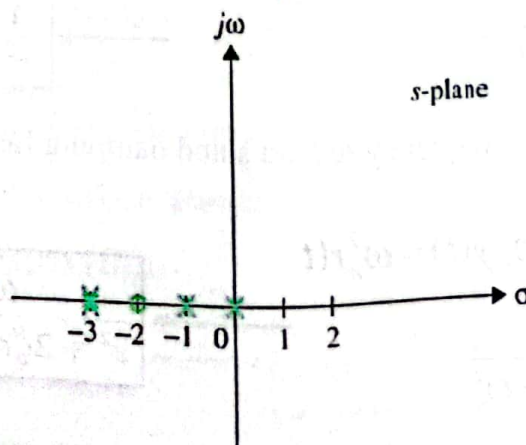
Poles and Zeros of Transfer Function

- **Poles and Zeros:**

- Example:

$$G(s) = \frac{10(s+2)}{s(s+1)(s+3)^2}$$

- It has a pole of order 2 at $s = -3$ and simple poles at $s = 0$ and $s = -1$.
- It has one zero at $s = -2$.
- graphical representation of the finite poles and zero of the system in the s-plane.



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- **Poles and Zeros:**
- Example: When dealing with the time response of control systems, complex-conjugate poles (or zeros) play an important role, and as a result, They deserve special treatment here. Consider the transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- Let us assume that the value of ζ is less than 1, so that $G(s)$ has a pair of simple complex-conjugate poles a $s = s_1 = -\sigma + j\omega$ and $s = s_2 = -\sigma - j\omega$
- where $\sigma = \zeta\omega_n$ and $\omega = \omega_n\sqrt{1-\zeta^2}$

$$s = s_1 = Re^{j\phi} = R\angle\phi$$

$$s = s_2 = R(\cos\phi - j\sin\phi) = Re^{-j\phi} = R\angle-\phi$$

